

BhoomiScan Phase II: Hybrid Indic Document Reading and Deterministic Three-State Risk for Residential Due Diligence

OCR and translation as input; Python rules, cross-document compare, coverage, and a local SHA-256 chain as verdict

BhoomiScan | Your AIvocate. | Version 0.1.0 | Workspace BhoomiScanV18

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Builds on Phase I intake. Does not claim a public blockchain or a legal guarantee.

Abstract

In the real estate sector, ownership disputes, forged property papers, and weak transparency in land transactions remain common problems. BhoomiScan is a property-document verification platform that uses artificial intelligence, multilingual natural language processing, and blockchain-based hash records to make validation more reliable. The platform is designed to accept, assess, and verify property documents written in regional languages as well as in English. Optical character recognition and natural language processing extract key fields from the uploaded files, including owner names, survey numbers, registration details, and property descriptions. The extracted data are then compared across documents to flag inconsistencies, conflicting ownership details, and other irregular or fabricated-looking entries. Hashes of documents and verification records are stored on a blockchain-style ledger so that later tampering can be detected and the trail of checks remains auditable. The novelty of this work is not the use of OCR or hashing by themselves. It is a hybrid method in which document AI is used only to read text, while deterministic rules assign each risk category one of three states: DETECTED, NO_ISSUE_FOUND, or NOT_VERIFIED. Missing papers are never treated as a clean title. A source-text veto restores stay orders and case details if the extractor omits them. Survey numbers, names, and areas are compared across files with explicit MATCH or MISMATCH, not against a government portal. Verification coverage and the numeric risk score are kept as two different measures so completeness is not confused with safety. Integrity is a local SHA-256 GENESIS hash chain that is recomputed on verify; it is not a public blockchain. By supporting Indian-language papers and reducing purely manual reading, BhoomiScan aims to simplify property verification, lower the human effort required for a first-pass review, and reduce the chance that fraudulent land and real estate documents pass unnoticed.

Keywords: *property document verification, hybrid OCR and rules, three-state verification, cross-document comparison, SHA-256 hash chain, multilingual NLP*

I. Introduction

A stored PDF is not a reviewed title. Karnataka residential files are often scanned Kannada deeds mixed with English certificates [1], [2]. A buyer who cannot read the script, or who treats a missing encumbrance certificate as good news, is exposed. Generic chat systems compound the exposure: they can declare a property clear when the text is silent.

BhoomiScan Phase II starts from the Phase I vault (JWT user, residential case, validated bytes on disk) and adds a reading-and-verdict pipeline. The methodological split is the contribution: a commercial Indic document service is an input device; verdicts are Python functions that a unit test can freeze [3], [4].

A. Contributions

- 1) A hybrid pipeline: digitise and translate with Sarvam; classify, merge, score, and compare in Python.
- 2) A three-state risk language (DETECTED / NO_ISSUE_FOUND / NOT_VERIFIED) so missing evidence cannot become a green tick.
- 3) A source-text veto on stay, restriction, pending status, and case number so extraction JSON cannot hide litigation language present in OCR.
- 4) Deterministic cross-document comparison with survey-prefix normalization, name folding, and 2% area tolerance.
- 5) Separation of risk score (points only from DETECTED) from verification coverage (five-category completeness).
- 6) A local SHA-256 GENESIS chain over canonical report JSON, verified by recomputation, explicitly not a public blockchain.

B. What this paper refuses to claim

- That the property is legally clear, safe, or guaranteed.

- That the hash chain is Bitcoin, Ethereum, mining, a wallet, or a smart contract.
- That government portals are scraped or that payments and OAuth are live.
- That an LLM decides MATCH versus MISMATCH.

C. Organization

Section II places the work against manuals, portals, and opaque AI. Section III states the problem. Section IV is the methodology (algorithms and formulae). Section V is design. Section VI is implementation. Section VII is results. Section VIII concludes.

II. Related Work

A. Manual due diligence

Advocates already perform the five checks this product names: ownership, litigation, encumbrance, approval, and survey or khata consistency [1], [2]. Manual work does not automatically OCR Kannada, does not normalize Sy. No. against Survey No., and does not leave a recomputed hash of the written opinion. Phase II is a first pass for a buyer, not a substitute for that professional [5].

B. Document AI and Indic OCR

Optical character recognition for Indian scripts is a long-studied problem [6]. Vendor document-AI products now wrap OCR, layout, and optional structured extract. Sarvam is used here as such a vendor [3]. Using a vendor is not the novelty. The novelty is what is not delegated: status assignment, comparison, scoring, and integrity. Related extraction work that stops at JSON fields still leaves the buyer with a silent mortgage when no encumbrance paper was uploaded.

C. Risk scores and checklists

Binary safe/unsafe scores hide incompleteness. Software-requirements practice asks that unspecified behaviour be explicit [7]. Our three-state model is closer to a requirements checklist than to a credit score: NOT_VERIFIED is a first-class outcome. Coverage percent is reported separately so a filled vault is not confused with a low-risk vault.

D. Integrity of reports

SHA-256 is the NIST secure hash [8]. Hash chains appear in many audit logs. Public blockchains add consensus and tokens that this student system does not need and does not implement. We store previous_hash = GENESIS for the first record and link later versions by report version, not by wall-clock time, to avoid timezone false tamper. Related work on trusted timestamping (RFC 3161) is future work, not a claim [9].

E. Gap

No student-scale system we implement here combined Indic reading, a vetoed merge, three-state categories, pairwise field compare, dual meters (score versus coverage), and a recomputed local hash, while refusing to call silence safety. That combination is the Phase II gap.

Approach	Reads Kannada	Absence rule	Compare fields	Integrity
Manual checklist	Human only	Often implicit	By eye	None
Vendor OCR alone	Yes	Usually none	No	None
LLM chat over scans	Maybe	Can invent clear title	Unstable	None
Public blockchain stamp	No	No	No	Consensus tokens
BhoomiScan Phase II	Vendor + rules	NOT_VERIFIED default	Deterministic	Local SHA-256

Table 0a. Related approaches versus this paper. Public blockchain is a different problem.

III. Problem Statement

Given a Phase I case C with documents D1..Dk, produce a case-level view that is explainable and testable. Each Di may be Kannada, English, or mixed. The system must not conclude NO_ISSUE_FOUND merely because a category was not mentioned. It must notice when two files disagree on survey, names, area, or boundaries. It must let the buyer add an encumbrance certificate later without paying for OCR again. After finalize, a later reader must be able to tell VALID from TAMPERED without trusting the stored hash blindly.

Academic question: can Indic document AI plus deterministic rules present five categories as DETECTED, NOT_VERIFIED, or NO_ISSUE_FOUND with evidence, compare fields without an LLM, and seal a versioned report with SHA-256, without faking a public blockchain or a legal guarantee?

IV. Proposed Methodology

A. Hybrid stance

Fig. 1 separates I/O from verdict. Sarvam is invoked only on Analyze. Recalculate reads stored original_text, translated_text, and extracted_data. Tests monkeypatch digitise, translate, and extract so pytest never bills the vendor [4], [10].

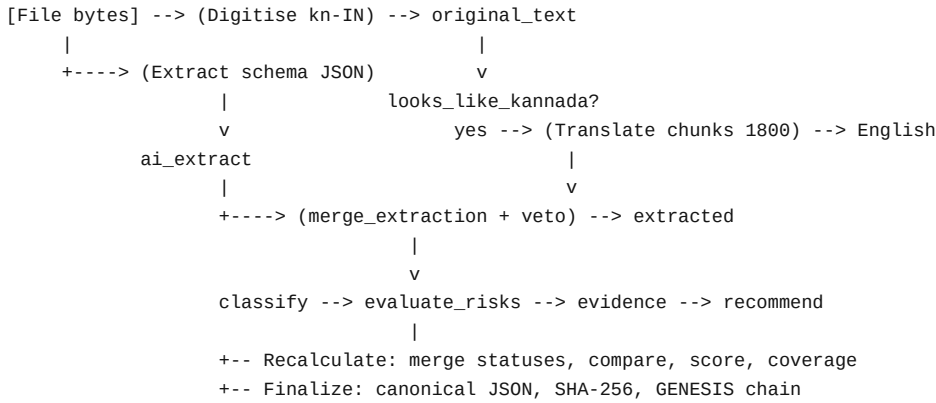


Fig. 1. Phase II control flow. The LLM path, if any inside the vendor, does not assign risk status.

B. Language detection and translation

Digitise is requested with language kn-IN because the target corpus is Karnataka residential paper. After OCR, let K be the count of Unicode letters in U+0C80 to U+0CFF and L the count of A-Za-z. The document is treated as Kannada if $K \geq 8$ or $K > L$ [6]. Kannada text is translated to en-IN with model sarvam-translate:v1 in chunks of 1800 characters. English-only files skip translation. Other Indian languages are not first-class in this phase; that limitation is stated, not hidden.

C. Classification

A keyword scorer on filename plus working text chooses among sale deed, parent deed, encumbrance certificate, khata, mutation, court papers, mortgage papers, building approvals, survey sketch, tax receipt, and others. Confidence below 0.45 maps to OTHER. A litigation-heavy sale deed may classify as a court-case document; tests and the interface must still show litigation if the text contains it.

D. Extraction merge and source-text veto

The extract schema groups property identifiers, ownership names, transaction fields, litigation, mortgage, approval, and four boundaries. Regular expressions supply fallbacks (survey/sy no, khata, area, executed-by / in-favour-of names, Kannada cues). Merge prefers a cleaned AI value when present, else the fallback. Veto (Algorithm 2): if fallback sees stay_order or transfer_restricted, those flags are forced true on the merged object; if fallback has a case number or pending status, they are restored when the model omitted them. This is the anti-hallucination rule in the opposite direction: the model is not allowed to forget a stay.

E. Three-state risk (Algorithm 3)

Categories are OWNERSHIP, LITIGATION, MORTGAGE, APPROVAL, PROPERTY_RECORD. Each receives one status and a level HIGH, MEDIUM, LOW, or INFO.

- DETECTED: explicit adverse evidence (incomplete seller/buyer names; pending case, stay, restriction; open charge without closure; unauthorized construction; later overlay of a high-severity mismatch).
- NO_ISSUE_FOUND: a relevant document explicitly clears the category (example: encumbrance certificate with nil / no / free from encumbrance).
- NOT_VERIFIED: default when papers are missing or silent. No EC means mortgage NOT_VERIFIED, never NO_ISSUE_FOUND.

Ownership is never auto-cleared from a survey number alone. A sale deed is not building permission. Lien matches on word

boundaries so alienation is not a mortgage. Case merge: DETECTED wins; else any NO_ISSUE_FOUND; else NOT_VERIFIED; then MISMATCH overlays DETECTED on the affected category.

Category	Typical DETECTED trigger	Typical NO_ISSUE_FOUND	Default
Ownership	Missing seller or buyer name	Never auto-cleared by survey only	NOT_VERIFIED
Litigation	Stay, pending O.S., restriction	Explicit no pending on a verify-type paper	NOT_VERIFIED
Mortgage	Open charge without closure	EC with nil / free from encumbrance	NOT_VERIFIED
Approval	Unauthorized or deviation	Approval paper plus present or OC	NOT_VERIFIED
Property record	Later MISMATCH on survey or bounds	Not granted from one identifier	NOT_VERIFIED

Table 0. Decision sketch for Algorithm 3. Defaults fail closed.

F. Comparison (Algorithm 4)

Fields compared include seller, buyer, owner, survey, khata, area, village, hobli, taluk, district, property type, north/south/east/west, registration number and date, and previous deed number. Normalization strips prefixes (Survey No., Sy. No., Khata No.). Names collapse whitespace and case. Area converts square metres to square feet by 10.7639 and allows 2% relative tolerance. If fewer than two documents carry a value, the result is NOT_AVAILABLE; otherwise MATCH or MISMATCH with severity and a plain-language explanation. No language model is consulted.

G. Score and coverage formulae

Let F be the set of findings after merge. Risk score uses only DETECTED rows:

$$\text{score} = \min(100, 70*|HIGH| + 15*|MEDIUM| + 10*|LOW|)$$

Overall level is HIGH if any DETECTED HIGH exists or score ≥ 40 ; MEDIUM if any DETECTED MEDIUM exists or score ≥ 10 ; otherwise LOW. NOT_VERIFIED adds zero points.

Coverage walks the five categories. Verified (DETECTED or NO_ISSUE_FOUND) weights 1.0. Partially Verified (NOT_VERIFIED but a related paper type was uploaded) weights 0.5. Not Verified weights 0.0.

$$\text{coverage\%} = \text{round}(100 * (\text{sum of five weights}) / 5)$$

An empty vault can have score 0 (LOW) and coverage 0%. That pair is intentional: low score is not safety; it may mean nothing was checked [7].

H. Evidence and recommendations

Each finding may store source snippets, page, document type, and field name. Recommendations list missing types (parent deed, EC, khata, mutation, court order, bank NOC, building approval, survey sketch) when a category is DETECTED or NOT_VERIFIED and the type is not already uploaded.

I. Local integrity method

Finalize builds report_content (property, documents, findings, evidence, mismatches, unverified, recommendations, disclaimer). report_hash = SHA-256 of JSON with sorted keys and compact separators [8], [11]. The first ledger row uses previous_hash = GENESIS. record_hash = SHA-256 of {report_id, report_hash, previous_hash, timestamp}. A later version links to the prior record_hash, selected by report version. Verify recomputes both hashes and the predecessor link. The on-disk CHAIN.json file states in plain English that it is not Bitcoin or Ethereum. A second finalize never edits the old row; it increments version.

- 1) Rebuild canonical JSON from stored report_content with sorted keys.
- 2) Compute h1 = SHA-256(canonical). If h1 $\neq$ report_hash, return TAMPERED.
- 3) Rebuild ledger material from report_id, report_hash, previous_hash, hashed_at.
- 4) Compute h2 = SHA-256(material). If h2 $\neq$ record_hash, return TAMPERED.
- 5) If version is first, previous_hash must be GENESIS; else it must equal the prior version's record_hash.
- 6) If all checks hold, return VALID. Never treat the stored hex string as proof by itself.

J. Evaluation method

Phase II and case-level behaviour are locked by pytest with mocked Sarvam [10]. Live UI tests require SARVAM_API_KEY and are not mixed into pytest. Legal language on screen is constrained: potential issue detected;

based on documents provided; could not be verified. Forbidden: completely safe; legally clear.

K. Worked score and coverage example

Suppose a case has one analyzed Kannada sale deed. Litigation is DETECTED HIGH (stay and pending O.S.). Ownership is DETECTED MEDIUM (seller or buyer name missing). Mortgage, approval, and property record are NOT_VERIFIED, and no EC, approval, or sketch has been uploaded. Then $|HIGH|=1$, $|MEDIUM|=1$, $|LOW|=0$, so score = $\min(100, 70+15) = 85$ and overall HIGH. Coverage weights are 1.0, 1.0, 0.0, 0.0, 0.0, so coverage% = 40. After an EC with explicit nil encumbrance is analyzed and the case is recalculated, mortgage may become NO_ISSUE_FOUND. Score stays 85 if litigation and ownership are unchanged, but coverage becomes 60%. That is the point of two meters: adding a clearing paper improves completeness without pretending the stay order disappeared.

L. Recalculate versus Analyze

Analyze is allowed to be slow and expensive. Recalculate must be local. The method therefore stores original_text, translated_text, detected_language, document_type, and extracted_data on the document row. Case merge reads those columns. If the first digitise was poor, the buyer must click Analyze again; Recalculate will not silently invent better OCR. This rule is part of methodology, not an implementation accident.

V. System Design

A. Architecture

Phase I layers remain. New routers expose analyze, analysis GET, recommendations, evidence, recalculate, due-diligence, comparisons, finalize, reports, verify, and PDF. Analyze timeout at the client is 180 seconds. Sarvam poll is 3 seconds with a 120 second cap.

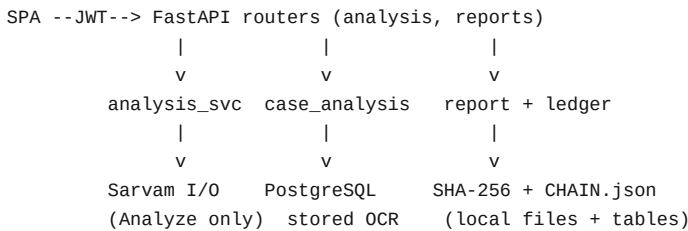


Fig. 2. Phase II architecture. Recalculate does not leave the database.

B. Data model extensions

Entity	New or used fields	Role
Document	original_text, translated_text, extracted_data, language, type	Stored reading
Analysis	status, pipeline_steps, extracted JSON	One-file run
RiskFinding	category, status, level, summary	Three-state row
EvidenceItem	source_text, page, field	Quote
Recommendation	category, document_type, reason	Missing paper
DocumentComparison	field, values, MATCH/MISMATCH	Pairwise
FinalReport	version, content, hash, score, coverage	Immutable version
LedgerRecord	previous_hash, record_hash, hashed_at	Local chain

Table I. Phase II entities. UUID keys and ownership checks from Phase I still apply.

C. Disclaimer design

Every stored report includes: This report is an AI-assisted document review and is based only on the documents provided. It does not replace advice from a qualified lawyer, surveyor, government authority, or other professional.

VI. Implementation

A. Modules

sarvam_service.py digitises, polls, reads page blocks, unpacks ZIP downloads, translates, and extracts. analysis_service.py is the one-file pipeline. extraction_service.py implements merge and veto. document_classifier.py scores types and

Kannada. risk_engine.py and recommendation_engine.py implement Algorithm 3. comparison_service.py implements Algorithm 4. scoring_service.py implements the formulae. hashing.py and ledger_service.py implement integrity. report_pdf.py writes the buyer PDF. NUL bytes are stripped before PostgreSQL TEXT.

B. API additions

Method	Path	Note
POST	.../documents/{doc}/analyze	May call Sarvam
GET	.../documents/{doc}/analysis	Latest one-file analysis
POST	.../{id}/recalculate	No Sarvam
GET	.../{id}/due-diligence	Same payload as recalculate
GET	.../{id}/comparisons	Stored pairs
POST	.../{id}/finalize	New version + ledger
GET	.../reports/{rid}/verify	VALID or TAMPERED
GET	.../reports/{rid}/pdf	Download

Table II. Phase II routes. All except health still require JWT and owner checks.

C. Frontend

Document vault offers Analyze and Recalculate. Analysis shows pipeline steps (uploaded, reading, extracting, checking, complete), extracted groups, and evidence. Due diligence shows overall level, score /100, coverage %, five cards, and the comparison table. Finalize warns that some checks may remain unverified and that missing documents do not block the report. The in-app chat page is a demo helper and is not the Sarvam path.

D. Implementation limits that are features

- Frontend never submits a trusted score; the server recomputes.
- Non-residential analysis remains gated as in Phase I.
- Subscription and social login remain non-functional screens.

VII. Results and Discussion

A. Automated results

Phase 2 tests (7) prove analyze requires auth, mocked OCR completes, 403/404 hold, explicit litigation becomes DETECTED, empty evidence does not become NO_ISSUE_FOUND, and recommendations appear. Case-level tests (19 in test_phase3.py, treated here as the Phase II diligence suite) prove name fallbacks, survey prefix MATCH, survey and area MISMATCH, recalculate merge, EC clearing mortgage, coverage, deterministic score, GENESIS linkage, second-version previous_hash, VALID on untouched JSON, TAMPERED on modified JSON, and 403 on a foreign report. Together with Phase I, the repository holds 39 test functions. Pytest does not call live Sarvam.

Observation	Rule confirmed
No EC uploaded	Mortgage stays NOT_VERIFIED
EC with nil encumbrance + recalculate	Mortgage may become NO_ISSUE_FOUND
Survey 45/3 vs 45/4	MISMATCH HIGH
Sy. No. 45/3 vs Survey No. 45/3	MATCH after normalize
Stay/O.S. language in OCR	Litigation DETECTED even if extract omitted it
Score only DETECTED HIGH	70 points, overall HIGH
All NOT_VERIFIED	Score 0, coverage may still be low
Edited report JSON	verify returns TAMPERED
Unedited finalize	verify returns VALID

Table III. Qualitative results from the mocked suite and the documented Kannada deed walkthrough.

B. Live walkthrough

With SARVAM_API_KEY set, a Kannada litigation sale deed is the main demonstration. Litigation is DETECTED at HIGH when stay, pending O.S., or transfer restriction appears. Mortgage and approval remain NOT_VERIFIED until supporting papers are analyzed. An early live failure (OCR download was a ZIP) was fixed by unpacking ZIP payloads in sarvam_service.py; that is an engineering result, not a change of method.

C. Why this is not a trivial wrapper

A wrapper would show vendor JSON and a green badge. Phase II adds a veto, a third status, a compare function with units, two different meters, and a recomputed chain. Those pieces are short in lines of code and long in specification, which is why they are written as algorithms and formulae rather than as a tool list [5], [7].

D. Consistency of terms

DETECTED, NO_ISSUE_FOUND, and NOT_VERIFIED are the only category statuses. MATCH, MISMATCH, and NOT_AVAILABLE are the only comparison results. Coverage uses Verified, Partially Verified, and Not Verified as labels on the five-category checklist; they are not a fourth risk status. Ledger means the local GENESIS hash chain in PostgreSQL and CHAIN.json. Blockchain, wallet, mining, and smart contract do not appear as implemented features. Phase I words (vault, case, buyer) keep the same meaning.

E. Threats to validity

Internal validity of the rule tests is high because Sarvam is mocked and the database is in-memory. External validity of live OCR is limited to scan quality and one vendor. Construct validity of risk is limited by design: the score is not a probability of losing title. Conclusion validity is protected by the disclaimer and by forbidding legally clear. Those threats are stated so the paper cannot be read as a larger study than it is.

F. Limitations

- OCR quality and vendor uptime bound live accuracy.
- Kannada party names still fail on some scans; incomplete ownership stays DETECTED.
- Boundary lines can concatenate during OCR.
- Hindi, Tamil, and other scripts are not first-class.
- The ledger is local; it is not a public timestamp authority [8], [9].
- No government fetch; no production hardening.

VIII. Conclusion

Phase II reads residential papers with Indic document AI and then refuses to let that AI decide risk. Five categories receive DETECTED, NO_ISSUE_FOUND, or NOT_VERIFIED. Silence stays unverified. Files are compared without an LLM. Score and coverage are not the same number. A finalized report can be shown VALID or TAMPERED by recomputing SHA-256 on a local GENESIS chain. The work does not claim a clean title and does not claim a public blockchain.

Future enhancement includes better Kannada name recognition, optional additional languages, lawful government APIs if they exist, RFC 3161 timestamps, and a human override with an audit note. None of those items are required to understand or reproduce the Phase II method described here.

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